

```
In [3]: import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

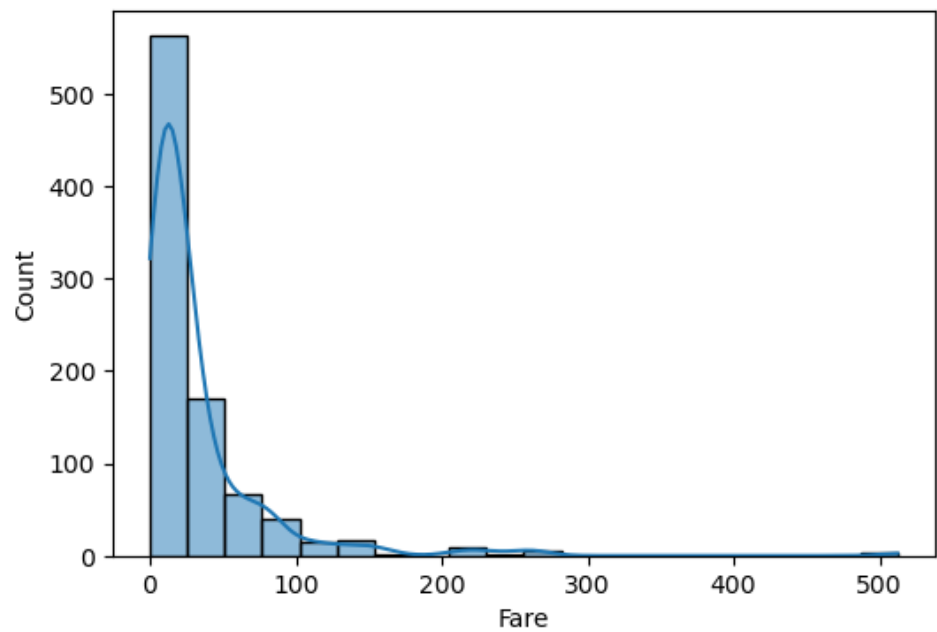
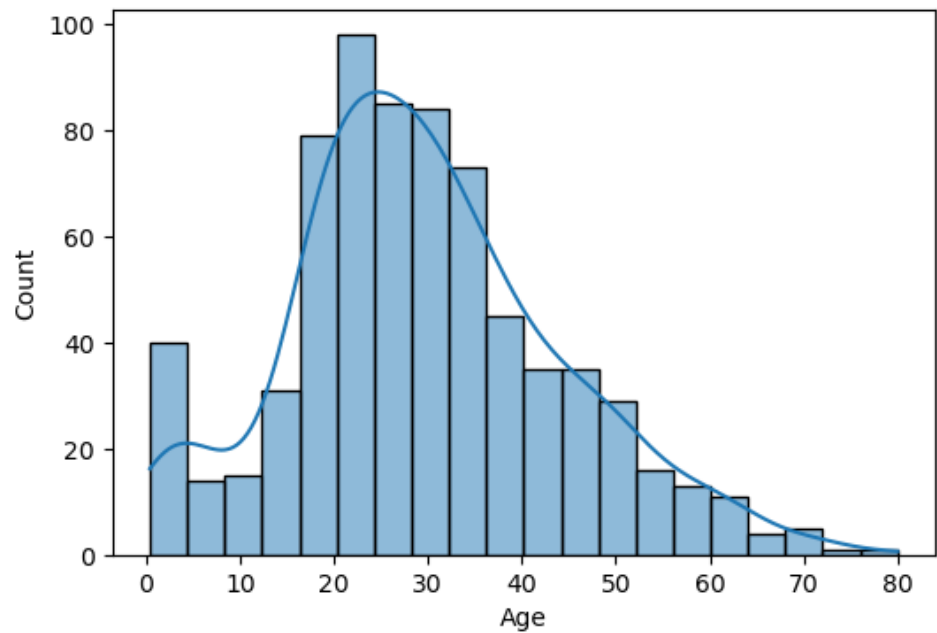
df = pd.read_csv("Titanic-Dataset.csv")
#Summary statistics(mean, median, std, etc.).
print(df.describe())
```

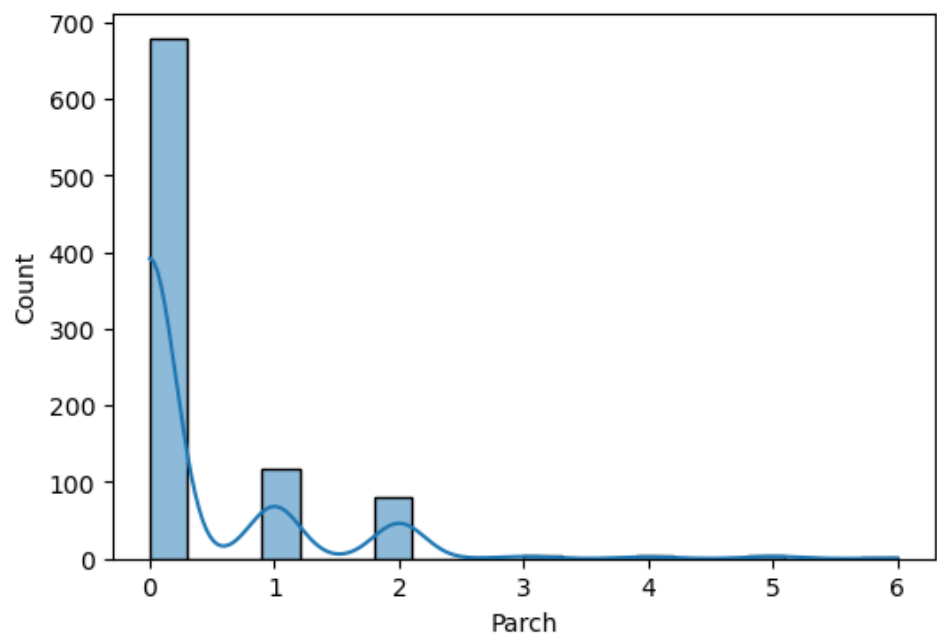
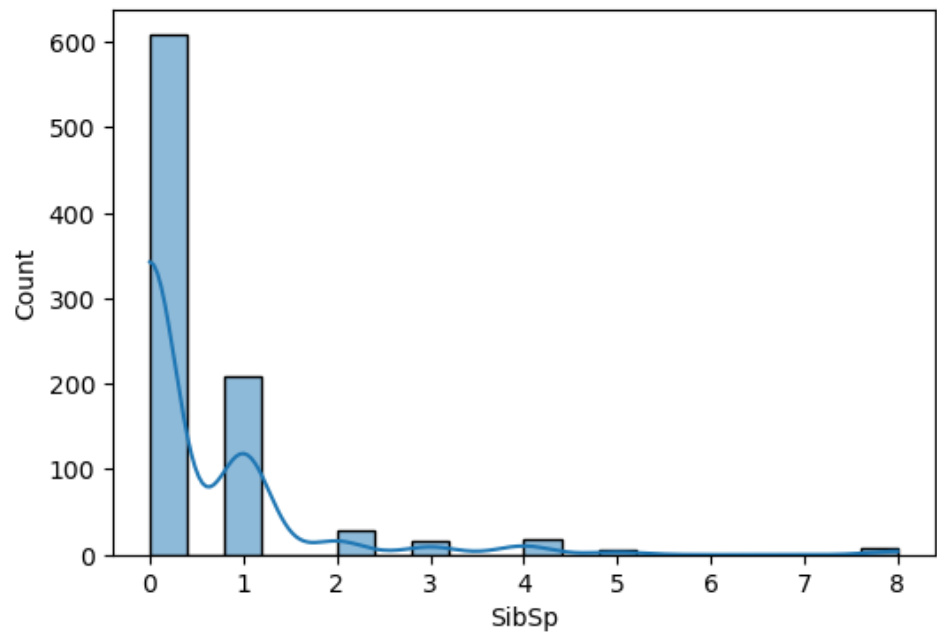
	PassengerId	Survived	Pclass	Age	SibSp
\					
count	891.000000	891.000000	891.000000	714.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

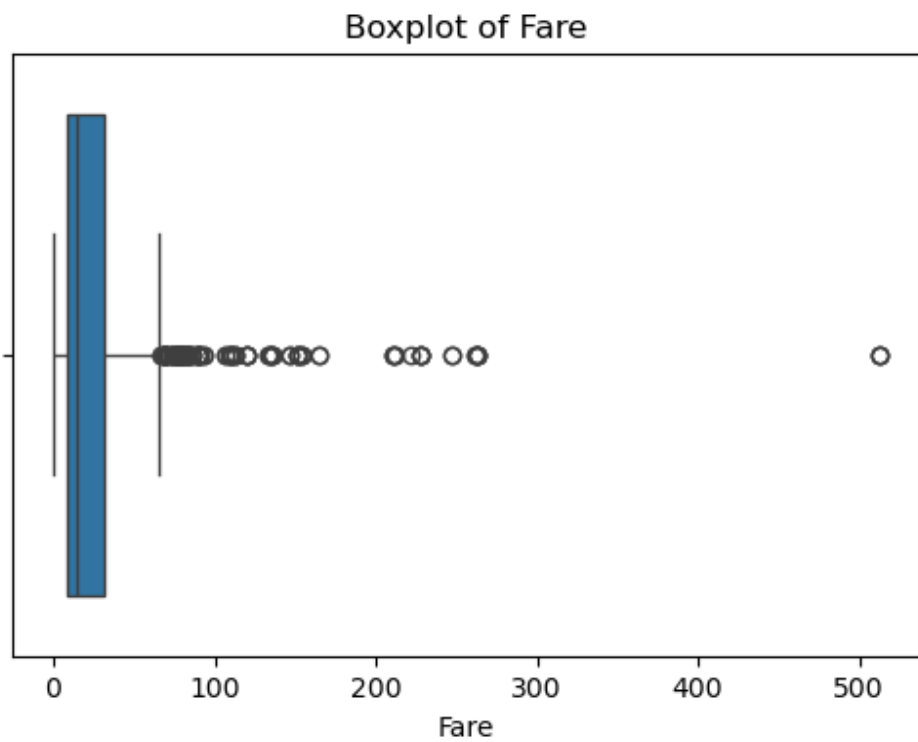
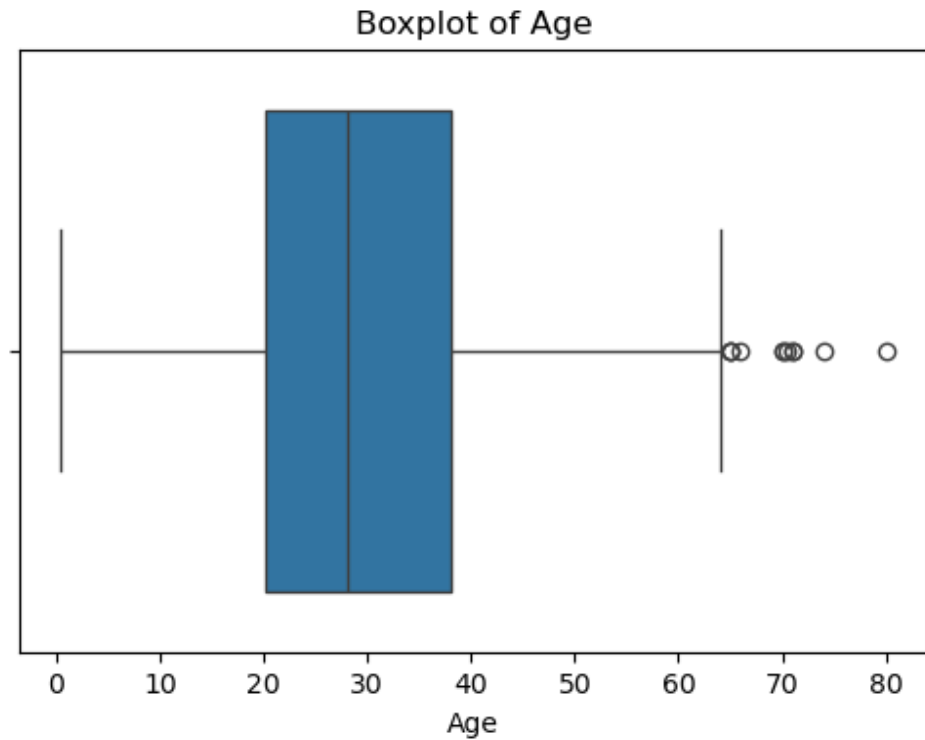
2. Histograms and Boxplots

```
In [9]: import seaborn as sns
num_features = ["Age", "Fare", "SibSp", "Parch"]
for col in num_features:
    plt.figure(figsize=(6,4))
    sns.histplot(df[col],bins=20,kde=True)
    plt.show()
```



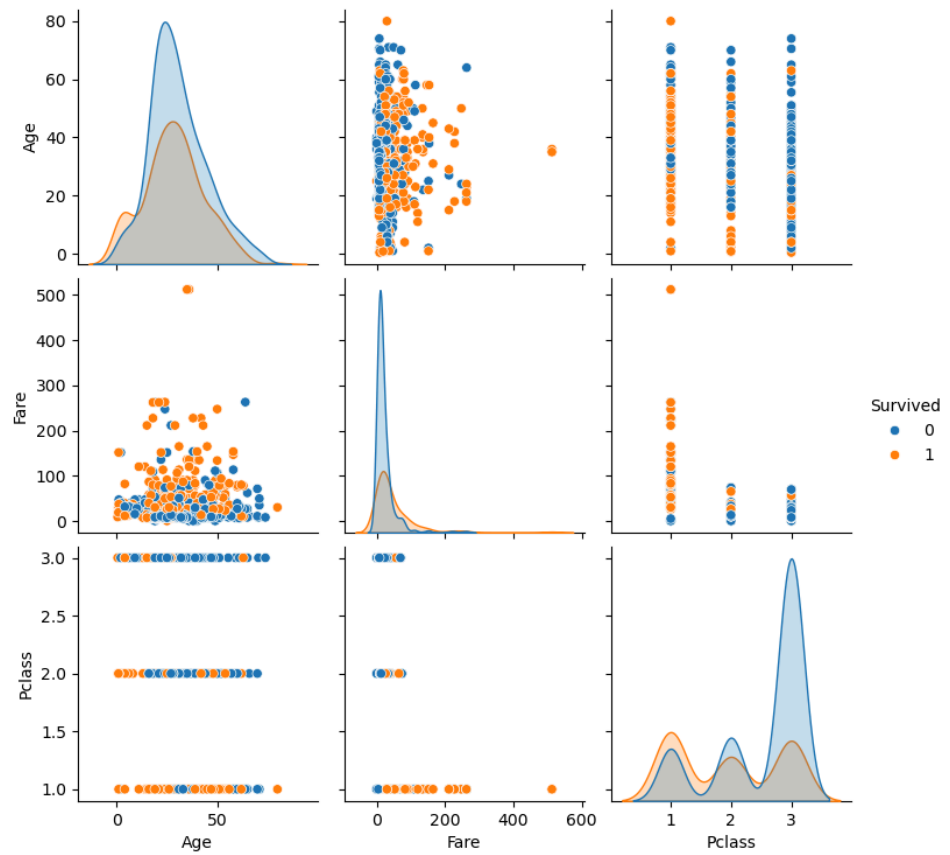


```
In [11]: #BOXPLOTS
box_features = ["Age", "Fare"]
for col in box_features:
    plt.figure(figsize=(6,4))
    sns.boxplot(x=df[col])
    plt.title(f'Boxplot of {col}')
    plt.show()
```

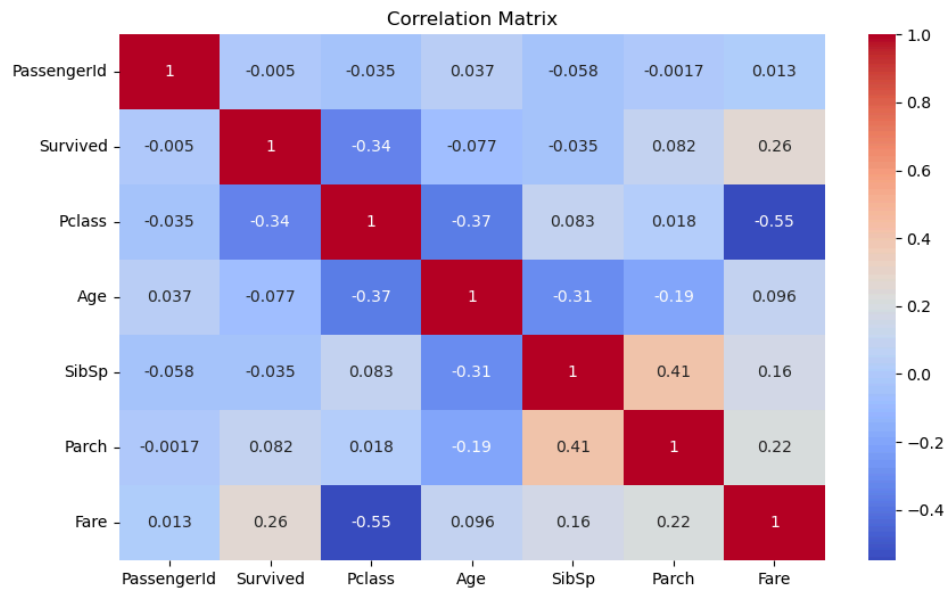


3. Pairplots and correlation matrix for feature relationships.

```
In [13]: #PAIRPLOTS
sns.pairplot(
    df[["Age", "Fare", "Pclass", "Survived"]],
    hue = "Survived"
)
plt.show();
```



```
In [14]: #CORRELATION MATRIX
corr = df.corr(numeric_only=True)
plt.figure(figsize=(10,6))
sns.heatmap(
    corr,
    annot=True,
    cmap="coolwarm"
)
plt.title("Correlation Matrix")
plt.show()
```



4. Identify patterns, trends, or anomalies in the data.

In []: